Exp: 11

Routing

11.a. Static Routing

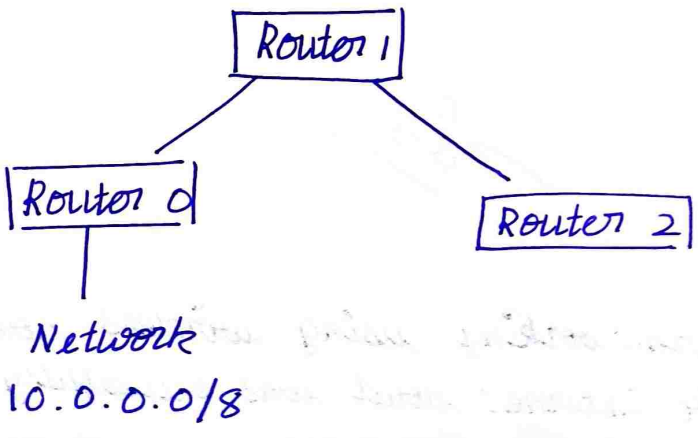
Aim:

To simulate and configure static routing using cisco packet tracer.

Observation:

1) Static Routing Involves manually adding routes to a router's Routing table.
2) Routers Automatically know the networks directly connected to their interfaces.
3) Routes are manually added for Networks that are available through other routers.
4) Both main & backup routes can be configured for a destination.
5) Administrative Distance (AD) is used to select the preferred static route.
6) The routing table is checked using show ip route static.
7) Connectivity and Routing paths are verified using ping & transcient.
8) When the main route fails, the backup route can be used for forwarding paths.

Diagram:



Experiment 11:

Router1

Physical Config CLI Attributes

IOS Command Line Interface

```
Router>en
Router#conf term
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#do show ip interface
GigabitEthernet0/0 is administratively down, line protocol is down (disabled)
  Internet protocol processing disabled
GigabitEthernet0/1 is administratively down, line protocol is down (disabled)
  Internet protocol processing disabled
GigabitEthernet0/2 is administratively down, line protocol is down (disabled)
  Internet protocol processing disabled
Vlan1 is administratively down, line protocol is down
  Internet protocol processing disabled

Router(config)#int g0/0
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

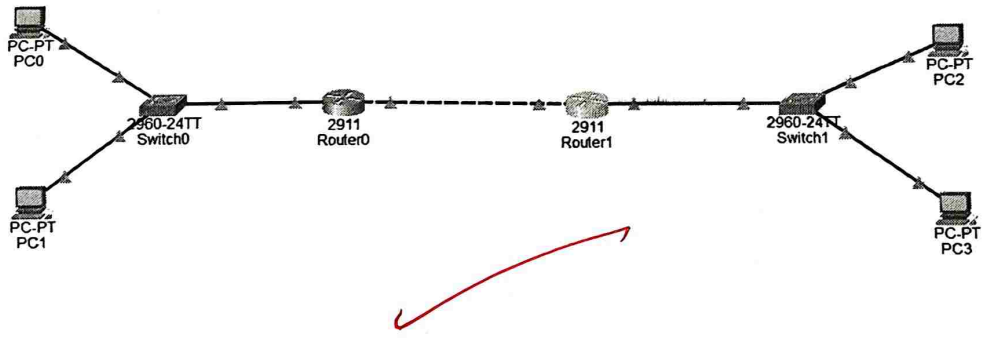
Router(config-if)#exit
Router(config)#interface g0/1
Router(config-if)#ip address 10.0.0.2 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

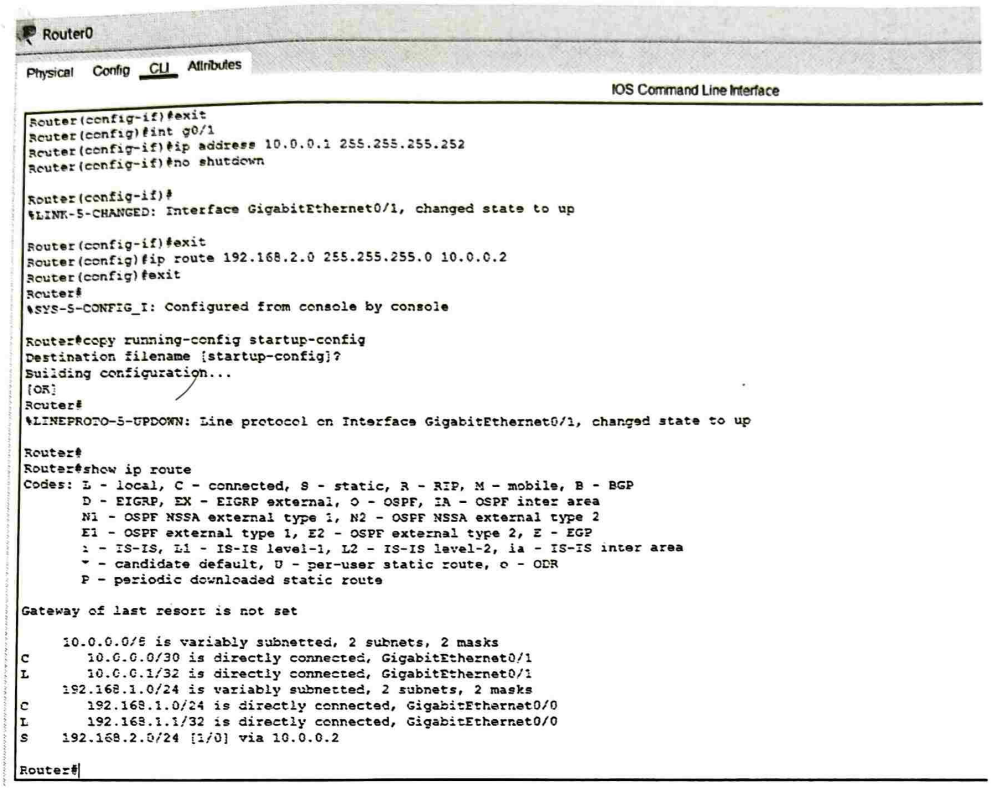
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router(config-if)#exit
Router(config)#ip route 192.168.1.0 255.255.255.0 10.0.0.1
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

```
Router#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
Router#
```



route when the main route



RIP_Protocol:

Result:

Thus, static routes were successfully configured and verified, including the use of a backup route when the main route becomes available.

11.b) RIP - Dynamic Routing

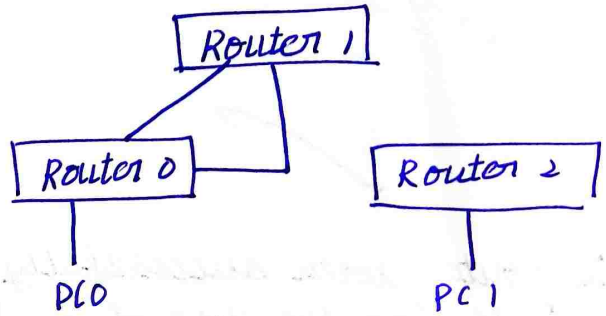
Aim:

To simulate and configure RIP(routing information protocol using CISCO Packet tracer

Observation:

1) Three routers are connected to form a network with multiple possible paths.
2) IP Addresses are assigned to the PC's, Fast Ethernet interfaces & serial interfaces.
3) RIP is enabled using vector rip command
4) Directly connected networks are advertised using network command.
5) RIP automatically exchanges routing information between Routers.
6) RIP selects the Route based on the lowest hop count.
7) The ping command is used to test connectivity between PC0 & PC1.
8) The transcart command is used to verify the path followed by packets.
9) If the selected Route fails, RIP automatically chooses another available route.

Diagram:



Router4

Physical Config CLI Attributes

IOS Command Line Interface

```
Press RETURN to get started!


Router>en
Router#conf term
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#do show ip interface
GigabitEthernet0/0 is administratively down, line protocol is down (disabled)
  Internet protocol processing disabled
GigabitEthernet0/1 is administratively down, line protocol is down (disabled)
  Internet protocol processing disabled
GigabitEthernet0/2 is administratively down, line protocol is down (disabled)
  Internet protocol processing disabled
Vlan1 is administratively down, line protocol is down
  Internet protocol processing disabled

Router(config)#interface g0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown

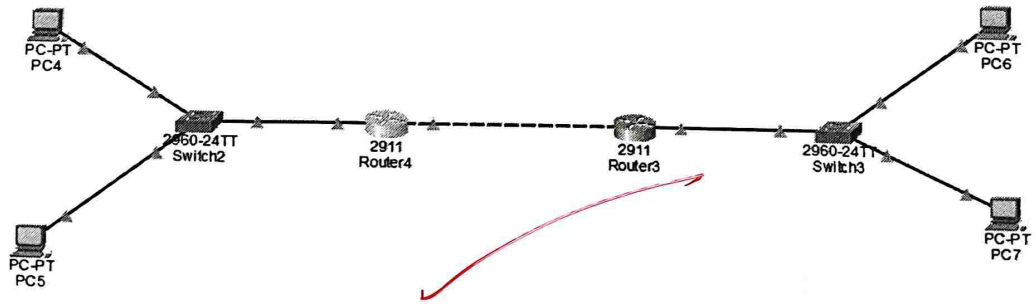
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

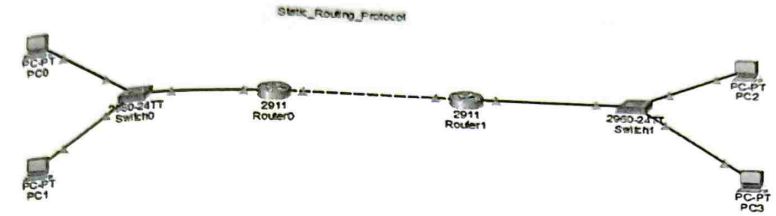
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

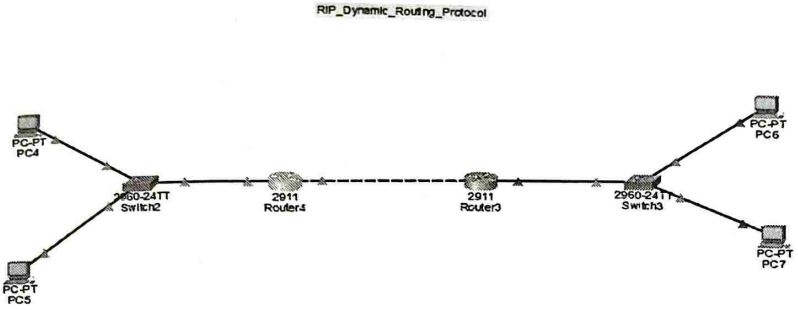
Router(config-if)#exit
Router(config)#route rip
Router(config-router)#version 2
Router(config-router)#no auto-summary
Router(config-router)#network 192.168.1.0
Router(config-router)#network 10.0.0.0
Router(config-router)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
Router#
```

RIP_Dynamic_Routing_Protocol




Static_Routing_Protocol
PC-PT
PC0
2960-24TT
Switch0
2911
Router0
2911
Router1
2960-24TT
Switch1
PC-PT
PC2
PC-PT
PC1
PC-PT
PC3


RIP_Dynamic_Routing_Protocol
PC-PT
PC4
2960-24TT
Switch2
2911
Router4
2911
Router3
2960-24TT
Switch3
PC-PT
PC6
PC-PT
PC7
PC-PT
PC5

Router 1:

PC0 → Router 0 → Router 1 → PC1

Router 2:

PC0 → Router 0 → Router 1 → Router 2 → PC1

Result:

Thus, RIP was successfully configured & dynamic Routing was verified.